Code

assignment6.py > ...

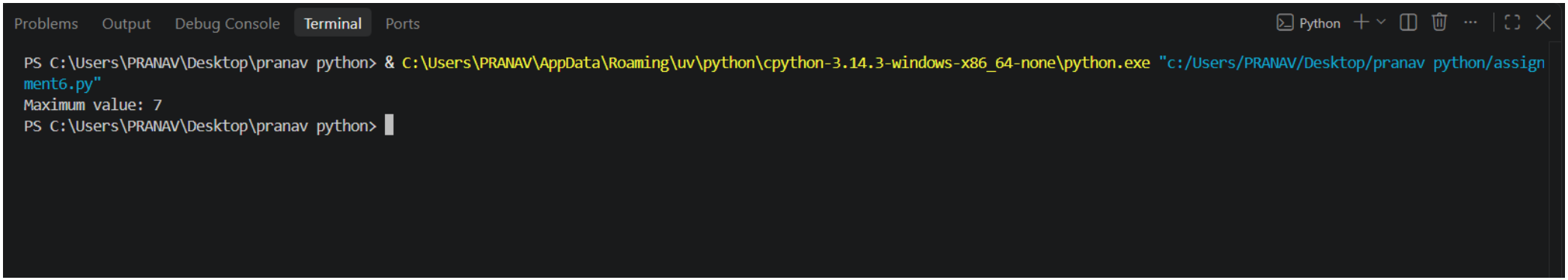
```python
def knapsack(weights, values, capacity):
    n = len(weights)
    dp = [[0] * (capacity + 1) for _ in range(n + 1)]

    for i in range(1, n + 1):
        for w in range(capacity + 1):
            if weights[i - 1] <= w:
                dp[i][w] = max(
                    values[i - 1] + dp[i - 1][w - weights[i - 1]],
                    dp[i - 1][w],
                )
            else:
                dp[i][w] = dp[i - 1][w]

    return dp[n][capacity]

def assignment6():
    weights = [2, 3, 4, 5]
    values = [3, 4, 5, 6]
    capacity = 5
    print("Maximum value:", knapsack(weights, values, capacity))
assignment6()
```

Output


Problems
Output
Debug Console
Terminal
Ports
Python
PS C:\Users\PRANAV\Desktop\pranav python> & C:\Users\PRANAV\AppData\Roaming\uv\python\cpython-3.14.3-windows-x86_64-none\python.exe "c:/Users/PRANAV/Desktop/pranav python/assignment6.py"
Maximum value: 7
PS C:\Users\PRANAV\Desktop\pranav python>